Meeting 5 SP23

# TensorDIMM

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# Outline

Background

Motivation

TensorDIMM

Evaluation

Section 1

# Background

# DRAM Architecture

This is a DIMM (Dual In-Line Memory Module)

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  - This is ineffective for server-class DRAM where the number of DIMMs can be large
  - Employ 'buffer' device per DIMM $\rightarrow$ intermediate controller
- Prior work has shown this buffer device can be used for custom logic
  - IBM CentaurDIMM

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- Growing trend towards dense, scaled-up system node design with multiple PCIe-attached co-processor devices
- Inter-device communication lies on the critical path of DL applications
  - Vendors are employing high-bandwidth interconnection fabrics with custom signaling links
- **System Goal:** Faster data transfers

# Embeddings

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- Embeddings are generally very sparse (information is not grouped together in memory).

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- Memory capacity limits of embedding layers
  - Embeddings scale with the number of items.
  - If a user is an item, a social media platform's embedding table can exceed several hundred GB

Section 2

# Motivation

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- GPUs often employ high-bandwidth memory (eg. GDDR, HBM)
- HBM is often capacity-limited when compared to CPU DRAM
- Technological constraints to adding more RAM:
  - Stacking DRAM vertically is constrained by chip pinout required to drive the added DRAM stacks, its wireability on top of silicon interposers, and thermal constraints
  - GPUs are close to the limits of processor die size so adding more 3D stack modules sacrifices budget for compute

# Memory Limits in Recommender Systems - Approach 1

*Model size is in the 100s of GB, much greater than GPU capacity*

- **CPU-only**: Run model on CPU. Gets around GPU memory bottleneck, but slow!

# Memory Limits in Recommender Systems - Approach 2

*Model size is in the 100s of GB, much greater than GPU capacity*

- **CPU-GPU hybrid**: CPU ‘memcpy‘ embeddings over to the GPU, then the GPU computes the output tensors and does the remainder of the recommender calculation. This bloats up memory bandwidth!

# Memory Limits in Recommender Systems - Figure

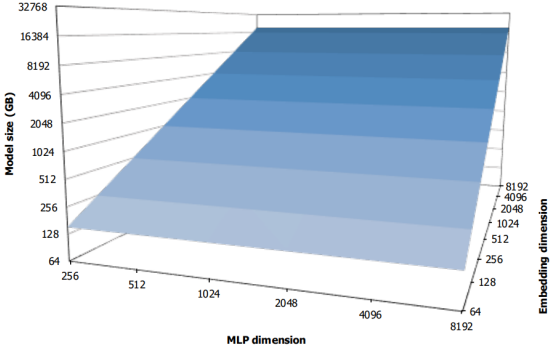


**Figure 3: Model size growth of neural collaborative filtering (NCF [26]) based recommender system when the MLP layer dimension size (x-axis) and the embedding vector dimension size (y-axis) is scaled up. Experiment assumes the embedding lookup table contains 5 million users and 5 million items per each lookup table. As shown, larger embeddings rather than larger MLP dimension size cause a much more dramatic increase in model size.**

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- CPU-only and CPU-GPU recommender systems need to trade-off the computational bottleneck of low-throughput CPUs over the communication bottleneck of PCIe
- **Urgent need for a system-level solution that provides scalable memory capacity and bandwidth expansion**

Section 3

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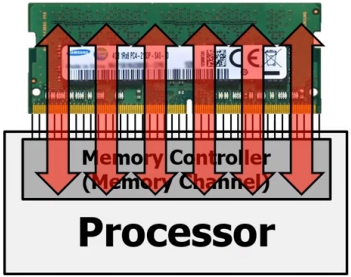
2. **TensorDIMM**, a practical near-memory processing architecture built on top of commodity DRAMs which offers scalable increase in both memory capacity and bandwidth for embedding gathers and tensor operations.
3. **TensorNode**, a TensorDIMM-based disaggregated memory system for DL. The efficiency of our solution is demonstrated as a proof-of-concept software prototype on a high-end GPU system, achieving significant performance improvements than conventional approaches.

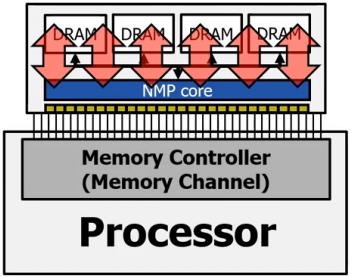
# Key Contributions

Embedding **gathers/reductions** are orchestrated between processor & DRAM




**Current system**

Embedding **gathers/reductions** are done **"locally"** within a DIMM




**TensorDIMM approach**

* Kwon et al., "TensorDIMM: A Practical Near-Memory Processing Architecture for Embeddings and Tensor Operations in Deep Learning", MICRO-2019

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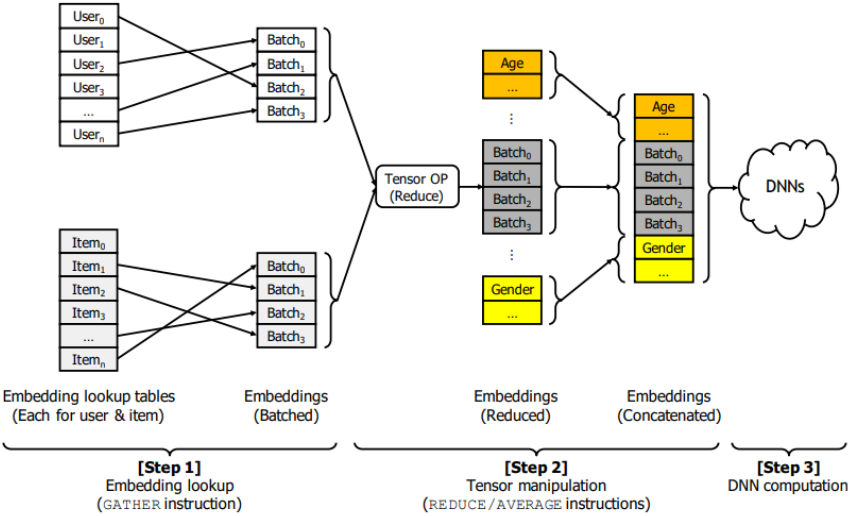
- Also functions as a normal DIMM if needed

# Vector ALU Functionality



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- 150 MHZ clock
- Negligable power + area overhead
  - IBM Centaur is a much more expensive design $\rightarrow$ TensorDIMM is commercially viable

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- Relies on purposeful address mapping to effectively utilize bandwidth
  - Consecutive 64B within an embedding vector are interleaved across different ranks

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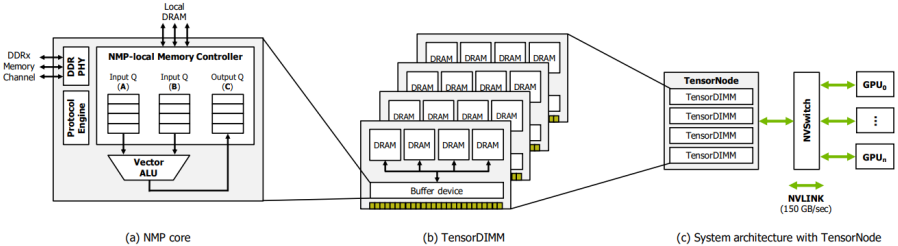
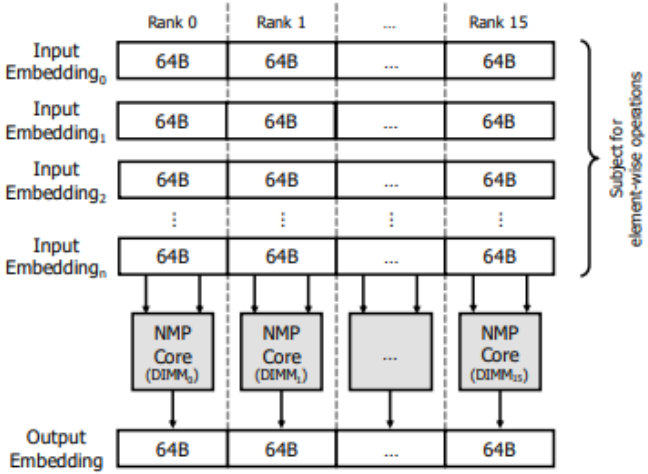



**Figure 6: High-level overview of our proposed system. (a) An NMP core is implemented inside the buffer device of each (b) `TensorDIMM`, multiples of which are employed as a (c) disaggregated memory pool, called `TensorNode`. The `TensorNode` is integrated inside the high-bandwidth GPU-side interconnect. The combination of NVLINK and NVSwitch enables GPUs to read/write data to/from `TensorNode` at a communication bandwidth up to 9× higher than PCIe.**

# Addressing Scheme



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  - Future-proof!

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- GPU runtime forwards CUDA kernel context to TensorNode network

Section 4

# Evaluation

# Evaluation Methodology

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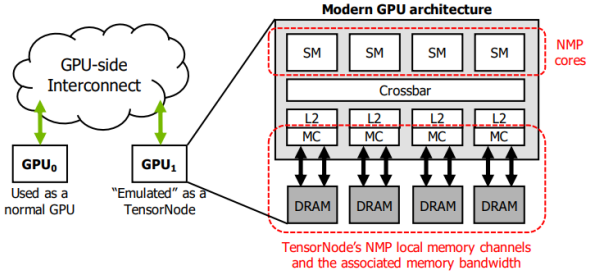
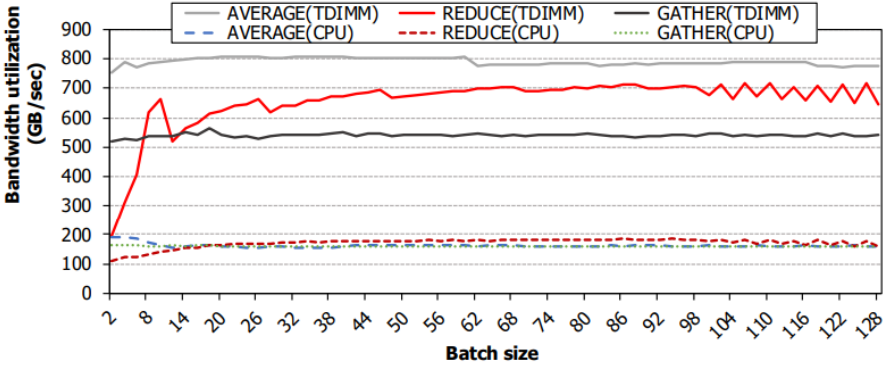
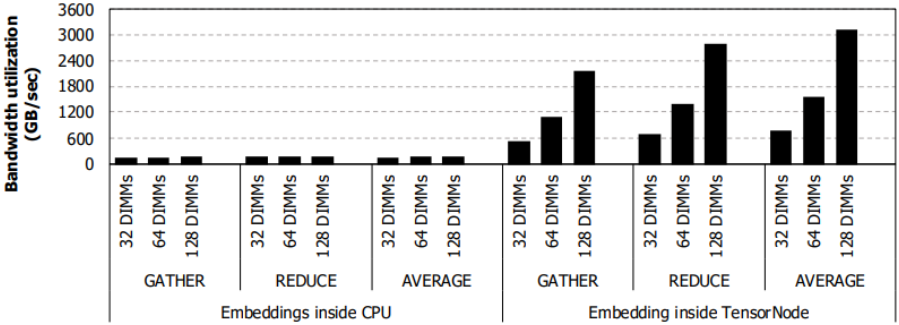


# Results


AVERAGE(TDIMM)
AVERAGE(CPU)
REDUCE(TDIMM)
REDUCE(CPU)
GATHER(TDIMM)
GATHER(CPU)
Bandwidth utilization (GB/sec)
900
800
700
600
500
400
300
200
100
0
2
8
14
20
26
32
38
44
50
56
62
68
74
80
86
92
98
104
110
116
122
128
Batch size

# Results


Bandwidth utilization (GB/sec)
3600
3000
2400
1800
1200
600
0
32 DIMMs
64 DIMMs
128 DIMMs
GATHER
REDUCE
AVERAGE
Embeddings inside CPU
Embedding inside TensorNode

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- Computing near memory can reduce bandwidth usage / requirements for embedding operations
  - DL workloads are computationally repetitive $\rightarrow$ ideal candidates
- **TensorDIMM makes embedding-focused DL workloads faster**

The end.